Manifold for T1D – Stewart-Lab Regeneration Analytics

DYRK1A-inhibitor, GLP-1 combination, and β-cell proliferation response prediction

The Stewart Lab's identification of **DYRK1A inhibition** as a reproducible inducer of human β-cell proliferation — and the super-additive proliferation observed under **DYRK1A inhibitor + GLP-1 receptor agonist** combination — has moved β-cell regeneration from biological aspiration to combinatorial design problem. Adding TGF-β superfamily inhibitors, glucokinase activators, and senolytics produces a compound-space that cannot be exhaustively screened. **Manifold** configures three engine capabilities to the Stewart-Lab data: causal discovery on the compound × donor × response matrix, criticality estimators (heterogeneous-treatment-effect learners, Gaussian-process regression on dose-response surfaces, change-point on proliferation markers) feeding a composite fragility score (**Ω**F), and Pearl do-calculus with interdiction for principled combination-therapy design on the DYRK1A signaling graph.

> Harmine-class DYRK1A inhibition has answered biology's question — human β-cells can be induced to proliferate pharmacologically. The outstanding question is which combination, in which patient, with what adjuncts sustains the expanded mass against continuing autoimmune pressure — a problem of causal inference.

ΩF PILLAR MAPPING – β-CELL REGENERATION PHYSIOLOGY

ΩF PILLAR	REGENERATION READING
I — Interaction	Compound × compound synergy across DYRK1A inhibitor + GLP-1 + TGF-β inhibitor combinations; donor genotype × compound response
R — Reserve	Baseline β-cell mass; post-treatment expanded mass via Ki67 / BrdU / EdU labeling index
J — Junction	Proliferation × apoptosis balance; autoimmunity × expanded-mass durability as translational concern
C — Cascade	DYRK1A → NFAT → cell-cycle-progression cascade; p16 senescence-induction pathway
T — Temporal	Hazard of proliferation-response loss; durability of expanded mass; time-to-response classification

PROPOSED PARTNERSHIP

Partnership with a Stewart-Lab principal investigator at the Mount Sinai Diabetes, Obesity, and Metabolism Institute is sought for:

1. Retrospective compound × human-islet screen data access under Mount Sinai's existing Data Use Agreement.
2. Co-authorship on the heterogeneous-donor-response and combination-design benchmark manuscript (twelve to fourteen weeks from data access).
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, or NIH NIDDK for the Phase 2 translational arm.

Apex Analytica contributes the causal-inference engine, the **Ω**F framework, and the engineering team. Full brief: `M-T1D-02-MTSINAI`.